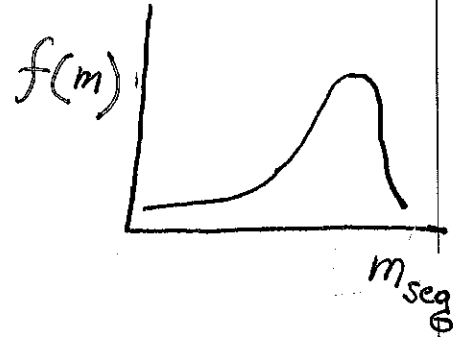


EMPIRICAL MAXIMUM LIKELIHOOD

⇒ Apply ML Techniques to NONPARAMETRIC DATA

e.g. CENTIPEDE SEGMENTS
(INTEGER RV)

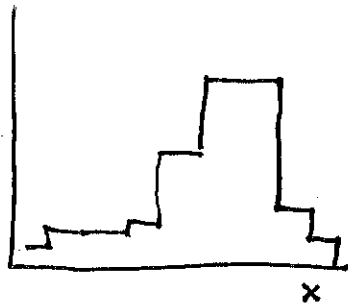


Q: CAN I DETERMINE THE MEAN (OR OTHER MOMENTS)
OF RV DISTRIBUTION FOR WHICH I DON'T HAVE AN
A PRIORI PARAMETRIZATION?

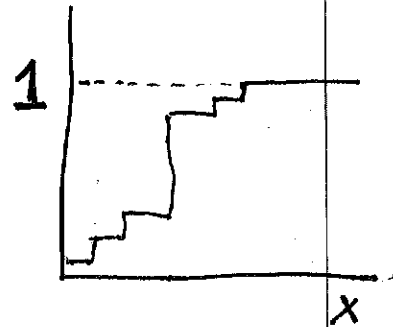
* * *

WHAT DO I KNOW FOR NONPARAMETRIC DATA?

EPDF
(Histogram,
Divide by
 $N \Delta X$)



ECDF



p_i = fraction of
all events
w/ in i 's bin

$$L = \prod_{i=1}^N p_i$$

$$\sum_{i=1}^N p_i = 1$$

WHAT IS
 L_{max} ? →

Consider $a + b = \gamma = \text{constant}$

↳ WHAT IS $\max a \cdot b$?

$$\frac{\partial}{\partial a} (a \cdot b) = \frac{\partial}{\partial a} (a(\gamma - a)) = 0$$

$$\gamma - 2a = 0$$

$$a = b = \frac{\gamma}{2}$$

- $a \cdot b$ maximum when $a = b$
- generalizes to any number of terms
- IN ECDF PLOT, THIS CORRESPONDS TO ALL STEPS BEING EQUAL $\rightarrow$ UNIFORM DISTRIBUTION

$$\begin{aligned} L &= \prod_{i=1}^N p_i \\ \sum_{i=1}^N p_i &= 1 \\ L_{\max} &= \prod_{i=1}^N \frac{1}{N} \end{aligned}$$

WHAT ABOUT NON-UNIFORM DISTRIBUTIONS?

⇒ Suppose I'm TOLD THAT $\mu_0 = \langle x \rangle$ is mean of data.
THIS IS A CONSTRAINT ON MY SYSTEM OF EQUATIONS.

$$L = \prod_{i=1}^N p_i$$

$$\sum_{i=1}^N p_i = 1$$

$$\langle x \rangle = \sum_{i=1}^N p_i x_i = \mu_0$$

$$\sum_{i=1}^N p_i (x_i - \mu_0) = 0$$

$$g(x, \mu_0) = x - \mu_0$$

"ESTIMATING FUNCTION"

↳ THIS WILL VARY depending
ON WHAT MOMENT I
AM TRYING TO DETERMINE.

WE NEED TO ~~NOT~~ MAXIMIZE L AND GET $p_i (= \tilde{p}_i)$

SUBJECT TO A CONSTRAINT

$$\frac{L}{L_{\max}} = \frac{\prod \tilde{p}_i}{\prod \frac{1}{N}} = \prod N \tilde{p}_i$$

(LATER, we'll see advantages TO THIS APPROACH)

CONSTRAINED MAXIMIZATION ⇒ LAGRANGE MULTIPLIERS

$$\tilde{p}_i = \frac{1}{[1 + \lambda (x_i - \mu_0)]^N} \quad \lambda \equiv \text{LAGRANGE Multiplier}$$

⚡ CONSTRAINT EQUATION BECOMES (Solve to find λ)

$$\sum_{i=1}^N \frac{(x_i - \mu_0)}{[1 + \lambda (x_i - \mu_0)]^N} = \phi$$

CONSTANT

$$\frac{L}{L_{\max}} = \prod_i N \tilde{p}_i = \prod_i \frac{1}{[1 + \lambda (x_i - \mu_0)]^N}$$

$$\ln\left(\frac{L}{L_{\max}}\right) = - \sum_i \ln(1 + \lambda (x_i - \mu_0))$$

"IT HAS BEEN SHOWN" (OWEN, 1980s) THAT

$$W = -2 \ln\left(\frac{L}{L_{\max}}\right)$$

Converges to a χ^2 distribution in $N_{\text{df}} = \# \text{ constraints} (\nabla)$

↳ our goal is to minimize W

Evaluate W for each possible value of μ .

Find $W(\mu)$ at which 95% cutoff occurs!

< NOTEBOOK me1e - example.ipynb >